STA380

Week 8 Practice Version A

Question 2

Estimate

$$\theta = \int_{\mathbb{R}} \sin(x^2)\frac{\sqrt{2}}{3\sqrt{\pi}}\exp\left\{-\frac{x^2}{18}\right\}dx$$

using importance sampling, where the importance function is the pdf of the Normal distribution. Assume that you need $n = 10^4$ replicates. You may select any choice of μ and σ.

To generate a random sample from the Normal distribution, one could use the `rnorm(n, mean = 0, sd = 1)` function, where `n` represents the number of observations, `mean` represents the mean, and `sd` represents the standard deviation.

(a) [**4 points**] Write the algorithm using importance sampling. **Show your work.**

(b) [**2 points**] Write out the `R` code for the algorithm using importance sampling.

(a) $f(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$ $\quad 2\sigma^2 = 18 \Rightarrow \sigma^2 = 9 \Rightarrow \sigma = 3$

$N(\mu = 0, sd = 3) \Rightarrow f(y) = \frac{1}{3\sqrt{2\pi}} \cdot e^{-\frac{y^2}{18}}$

$$\theta = \int \cdots \cdot \frac{f(x)}{f(x)}\, dx.$$

$f_{Normal(0,3)}(x)$

$$= \int \sin(x^2)\cdot\frac{\sqrt{2}}{3\sqrt{\pi}}\cdot e^{(-\frac{x^2}{18})} \cdot \frac{\frac{1}{3\sqrt{2\pi}} e^{-\frac{x^2}{18}}}{\frac{1}{3\sqrt{2\pi}} e^{-\frac{x^2}{18}}}\, dx$$

$$\theta = \int 2\cdot\sin(x^2)\cdot f_{Normal(0,3)}(x)\, dx$$

$\frac{\sqrt{2}}{3\sqrt{\pi}} \div \frac{1}{3\sqrt{2\pi}}$

$\frac{\sqrt{2}}{\cancel{3}\sqrt{\pi}} \times \frac{\cancel{3}\sqrt{2}\cdot\cancel{\sqrt{\pi}}}{1} = 2$

Algorithm:

① Let $x_1, x_2 \cdots x_{10^4}$ be random variables generated from Normal (mean = 0, sd = 3).

② $\hat{\theta} = \frac{1}{10^4}\sum_{i=1}^{10^4} 2\cdot\sin(x^2)\cdot \mathbb{1}\{-\infty \le x \le \infty\}$

(b) R codes

```
n <- 10^4
x <- rnorm (n, mean =0, sd=3)
est <- mean (2 * sin (x^2))
```

Question 3 Practice Version A

Consider the left-truncated exponential distribution at T for $T > 0$:

$$f_X(x) = \frac{1}{\theta}\exp\left\{\frac{-x+T}{\theta}\right\}, \quad T < x < \infty.$$

(a) [**2 points**] What is the true value of $\int_T^{T+1} f_X(x)dx$?

(b) [**1 point**] Is it possible to use the antithetic approach to estimate $\int_T^{T+1} f_X(x)dx$? Why or why not?

(c) [**2 points**] Assuming it is possible, write the algorithm to estimate $\int_T^{T+1} f_X(x)dx$ using the antithetic variable approach. Let $n = 10^4$. Use any value of $\theta > 0$ that you desire.

(d) [**2 points**] Write out the R code for the algorithm using the antithetic variable approach.

(a) $$\int_T^{T+1} \frac{1}{\theta}\exp\left\{\frac{\overset{-(X-T)}{-x+T}}{\theta}\right\}, \quad T < x < \infty$$

$Exp(\theta)$

$f(y) = \frac{1}{\theta}\cdot\exp\{-\frac{x}{\theta}\}$

$$= \left(-\exp\left\{\frac{-x+T}{\theta}\right\}\right)'\Big|_T^{T+1}$$

$$= \cancel{-}\exp\left\{\frac{-x+T}{\theta}\right\}\cdot\cancel{-}\frac{1}{\theta}$$

$$= -\exp\left\{\frac{-(T+1)+T}{\theta}\right\} - \left(-\exp\left\{\frac{-T+T}{\theta}\right\}\right)$$

$$= 1 - \exp\left\{\frac{1}{\theta}\right\}$$

(b) To estimate $\int_T^{T+1} f_X(x)\,dx$. $\int_T^{T+1} \underset{1}{\cancel{\frac{1}{b-a}}}\frac{1}{\theta}\exp\left\{\frac{-x+T}{\theta}\right\}$

$$E[g(U)] = E_{U(T,T+1)}\left[\frac{1}{\theta}\exp\left\{\frac{-x+T}{\theta}\right\}\right]$$

the density we know $Y = X - T \sim Exp(\theta)$

when $X = T+1 \Rightarrow Y = T+1-T = 1$

$X = T \Rightarrow Y = T - T = 0$

$$\Rightarrow \int_0^1 \frac{1}{\theta} \exp\left\{\frac{-y}{\theta}\right\} dy$$

$Y = X - T$

$dy = dx$

$$= E_{U(0,1)}\left[\frac{1}{\theta}\exp\left\{\frac{-y}{\theta}\right\}\right]$$

Yes

↓ check monotone ✓

(c) Algorithm: ① Generate $U_i \overset{iid}{\sim}$ Uniform $(0,1)$ for $i = 1, 2, \ldots, \frac{n}{2}$

② Compute $\hat{\theta} = \frac{2}{n} \sum_{i=1}^{\frac{n}{2}} \left[\frac{\frac{1}{\theta} e^{\left\{-\frac{u_i}{\theta}\right\}} + \frac{1}{\theta} e^{\left\{-\frac{(1-u_i)}{\theta}\right\}}}{2}\right]$

$n = 10^4$

(d) R codes.

```
n <- 10^4
theta <- 1
U <- runif( n/2 )

g1 <- 1/θ * exp(- U/theta )

g2 <- 1/θ * exp (-(1-U) /theta )

est <- mean ( (g1+g2) /2 )
```

Question 5 Practice Version B

Let $X_1, X_2, \ldots, X_{30}$ be a random sample from the *Exponential*(θ) distribution. The goal of this question is to simulate a hypothesis test. Consider the hypothesis test:

scale check

$$H_0 : \theta = 3 \quad \text{v.s.} \quad H_1 : \theta < 3$$

TUT 7. Q2.

Someone begins simulating values using the following code:

```
n <- 30
alpha <- 0.05
theta <- 2
m <- 10^4
x <- matrix(rexp(n*m, rate = 1/theta), nrow = m)
```

H_α

The same person found the following **rejection region** (RR):

$$RR := \left\{ \sum_{i=1}^{30} X_i < k \right\}$$

(a) [**1 point**] Using R, find what the critical value k should be given that the significance level is 0.05. *Hint:* the formula sheet includes `qexp()`, `qgamma()`, and `qbeta()`. Potentially one or more could be useful.

(b) [**1 point**] Using R and the code present in the question, compute the test statistic. *Hint:* any one of these could be useful: `colSums()`, `rowSums()`, `colMeans()`, `rowMeans()`.

(c) [**1 point**] Using R, compute the Monte Carlo type-I error rate.

(a) $X_i \sim Exp(\theta)$

under H_0

$H_0 : \theta = 3$

$S = \sum_{i=1}^{30} X_i \sim Gamma(shape = n \leftarrow 30, scale = \theta \leftarrow 3)$

$\alpha = \text{Type I error} = P(RR \mid H_0)$

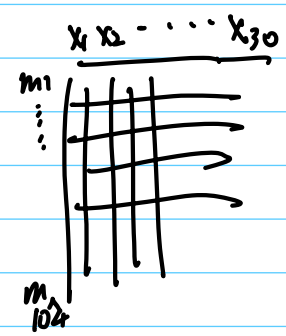
$= P(S < k) = 0.05$

$k \leftarrow qgamma(0.05, shape = 30, scale = 3, lower.tail = TRUE)$

$k \approx 64.78194$

(b)

```
n <- 30
alpha <- 0.05
theta <- 2          # under Hα
m <- 10^4
x <- matrix(rexp(n*m, rate = 1/theta, nrow = m))
```



rowSums(x)

(c) MC Type-I error rate.

```
n <- 30
alpha <- 0.05
theta0 <- 3          # under H0
m <- 10^4

x <- matrix (rexp(n*m, rate=1/theta0, nrow=m)

S <- rowSums(x)

k <- qgamma (alpha, shape=n, scale=theta0, lower.tail=TRUE)

typeI_error <- mean (S<k)
```

0.0548

Exp: $Y_1 \cdots Y_8$. r.s

$$f(x|\beta) = \begin{cases} \frac{3}{\beta} e^{-\frac{y^3}{\beta}} & , y>0 \\ 0 & , ow. \end{cases}$$

$H_0: \beta = 2$ v.s $H_a: \beta > 2$

$\alpha = 0.05$

RR : $\sum_{i=1}^{8} y_i^3 > k^*$

Review. $\sum_{i=1}^{8} Y_i^3 \sim$ Gamma $(8, \beta)$.

and if $\beta = 2$, then Gamma $(8, 2) = \chi^2(16)$

Recall: $\chi^2(n) = \text{Gamma}(n/2, 2)$